



BAC100 / BAC100B / BAC102 Lab Grade Raman Probe User Manual

Table of Contents

1	Fiber Optic Raman Probe Overview	3
1.1	Laser Warning Labels	3
1.2	Beam Shutter Switch	3
1.3	Warranty Seal Label	4
2	Diagrams and Dimensions of Raman Probes	4
2.1	BAC100 and BAC100B Probe	4
2.2	BAC102 Probe	5
3	Connecting Probe to Spectrometer	6
4	Operation	8
5	Maintenance	10
5.1	Exterior	10
5.2	Fiber Optic Tips	11

1 Fiber Optic Raman Probe Overview

The BAC100, BAC100B, and BAC102 are fiber optic lab-grade Raman probes to be used with B&W Tek portable Raman systems. The innovative probes use state-of-the-art package design and optimized optical lenses.

The fiber optic probe family contains two different products:

- BAC100, BAC100B Lab grade Raman probe without hand trigger
- BAC102 – Lab grade Raman probe with hand trigger.

The standard probe shaft is constructed from 304SS and designed for laboratory use. It is not for immersive use. This shaft is replaceable and optional shafts for use in solutions, and in harsher environments are available.

1.1 Laser Warning Labels

Located at the laser emission aperture, the laser emission label indicates that the laser energy emission occurs at the corresponding port.



1.2 Beam Shutter Switch

A manual beam shutter is located on the fiber optic probe. When the switch is set to “Open”, the beam shutter is open. When the switch is set to “Close” the beam shutter is closed.



1.3 Warranty Seal Label

The warranty seal label is positioned on the front side of the probe body. Any attempt to open the surface panel will break the seal and void the warranty.

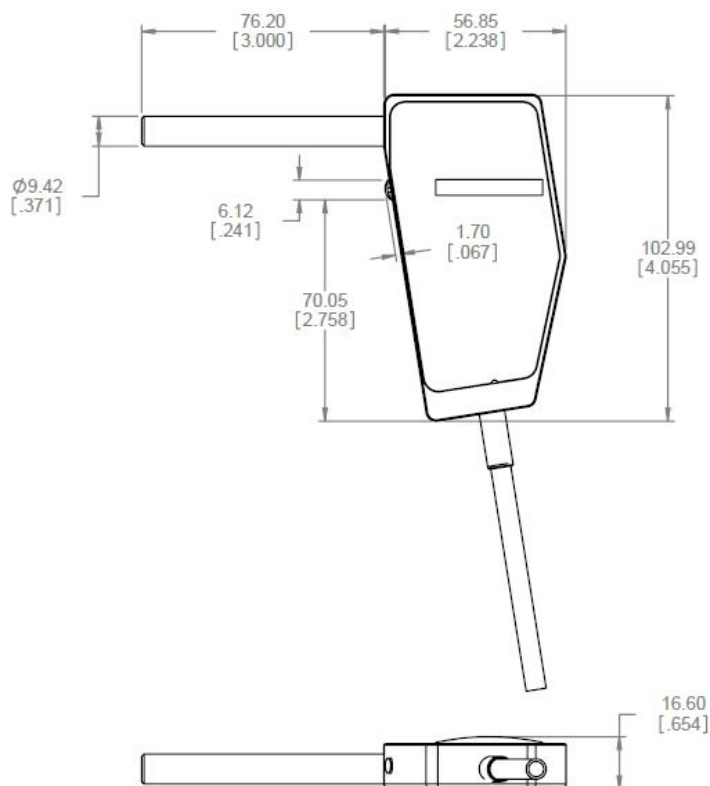


2 Diagrams and Dimensions of Raman Probes

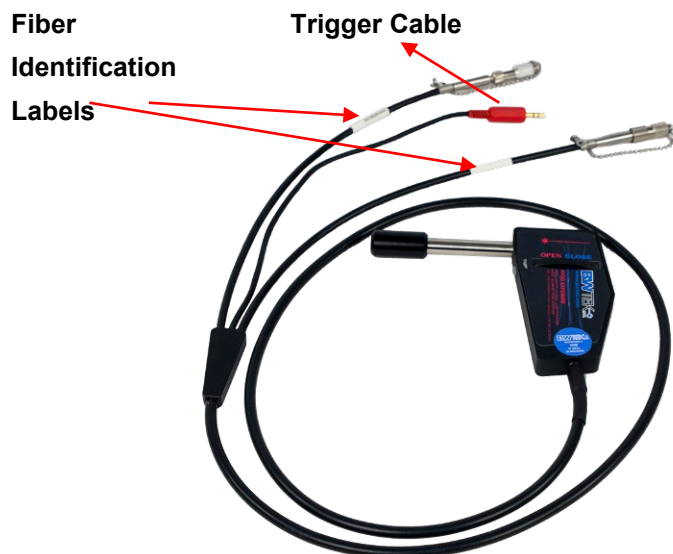
These Raman probes are equipped with a 3" (76.2 mm) with diameter of 0.371" (9.42 mm). The standard fiber length is 1.5 meters.

2.1 BAC100 and BAC100B Probe





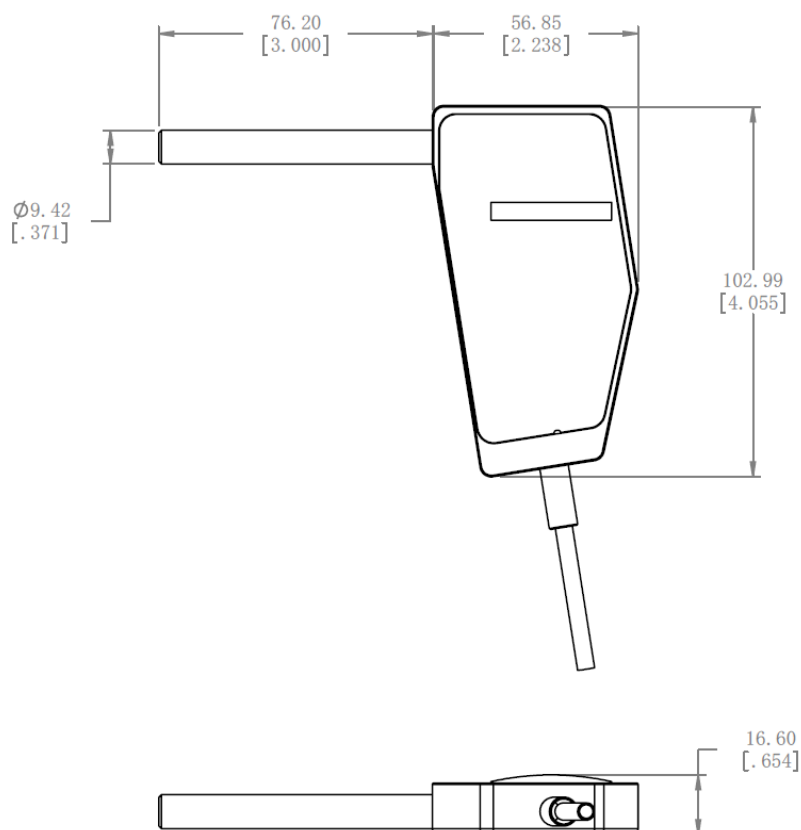
2.2 BAC102 Probe



Distance Regulator



Trigger Button



3 Connecting Probe to Spectrometer

- Check the identification label on each fiber. Connect the fiber labeled “Excitation” to the laser excitation port on the Raman system. Note that the excitation fiber tip is keyed and must be aligned to the notch at the excitation port in order to connect correctly. Push the fiber end into the port and tighten the screw.



Keyed Excitation Fiber End



Excitation Port with Notch

- Connect the fiber labeled “Collection” to the collection port on the Raman system.
Note: There are two types of collection fiber tip: with SMA connector and with FC/PC connector.
Different models of Raman system may have different types of collection port.

For collection fiber with SMA connector:

Push the fiber end into the SMA collection port and tighten the screw.



Collection Fiber End with SMA



Collection Port with SMA

For collection fiber with FC/PC connector:

Align the keyed collection fiber tip to the notch at the collection port. Push the fiber end into the port and tighten the screw.



Keyed Collection Fiber End



Collection Port with Notch

- For the BAC102 probe, connect the trigger plug by simply plugging the trigger cable into the “Trigger” port on the Raman system.



Trigger Port



BAC102 probe installed on an *i*-Raman Plus system

4 Operation

Warning: Although the probe by itself is not a laser product, when used with a Raman spectrometer, the system is a laser product, and safety guidelines provided in the instrument's user manual must be followed.

The fiber optic Raman probe is easy to use. Each probe has a manual shutter to close/open the laser beam. Ensure that the probe shutter is open when during measurement, and closed when the probe is not in use.

Optimum Raman signal is obtained when the sample is placed at the focal point of the shaft. The working distance of the standard shaft, defined as the distance from the focal point in air to the end surface of the shaft, is 5.4 mm, 5.5 mm and 5.9 mm for 532 nm, 785 nm and 1064 nm models, respectively.

For measurement of liquids in transparent glass containers, the shaft can be placed in contact with the container. The beam can be focused in the liquid through a glass wall up to 5 mm thick.

For measurement of solid material, the distance regulator can be used. Slide the distance regulator onto the shaft until it can go no further, and the focal point will be slightly outside the regulator surface. Measurement can be made easily by place the probe with the distance regulator in direct contact with the solid sample.

A variety of sampling accessories are available, including.

- Liquid Sample Cuvette Holder (BCR100A)
- Liquid Vial holder (NR-LVH)
- Raman Flow Cell (BAC160-SS)
- Probe holder & XYZ positioning stage (BAC151B)
- Video microscope (BAC151C)

The shaft is screwed into the probe head and is user replaceable. Several types of replaceable shaft are available for different applications, for example,

- 12.0 mm diameter immersion shafts (RIS100-FS and RIS100-SA)
- High pressure and high temperature immersion shafts (RIS101-SS and RIS101-HS).
- Hot melt extrusion shafts coupled to standard Dynisco ½-20 UNF ports (RHS100-SSG-08, RHS100-SSE-08).

- A range of long-working-distance lenses for standoff measurements.

Warning: Never operate the probe connected to a laser without a shaft attached. Doing so will cause a collimated laser beam to exit the probe, which will greatly increase the NOHD and the chance of eye injury.

5 Maintenance

The fiber probe is an optical instrument and proper maintenance is important to ensure a long service life.

There are no user serviceable parts inside the probe head. Do not allow anyone except trained Metrohm engineer to open the probe housing.

Warning: Do not turn on the laser unless the laser port is covered with the chained metallic dust cap, or plugged in by the probe's laser delivery fiber optic cable. The laser port must be covered by the metallic dust cap or plugged by the probe excitation fiber all the time during operation and maintenance. If laser is turned on, an uncovered laser port could result in direct eye or skin exposure to direct laser radiation.

Fiber optic probe care is important for normal operation of the instrument. The fiber tips are optically polished and inspected by a 200x microscope before shipping. Microscopic foreign materials such as dust particles, debris and tissue on the excitation fiber end may burn under high intensity laser power. Should burning on the fiber tip occur, it may lead to a significant drop in laser output power, and the fiber must be re-polished to fix the problem.

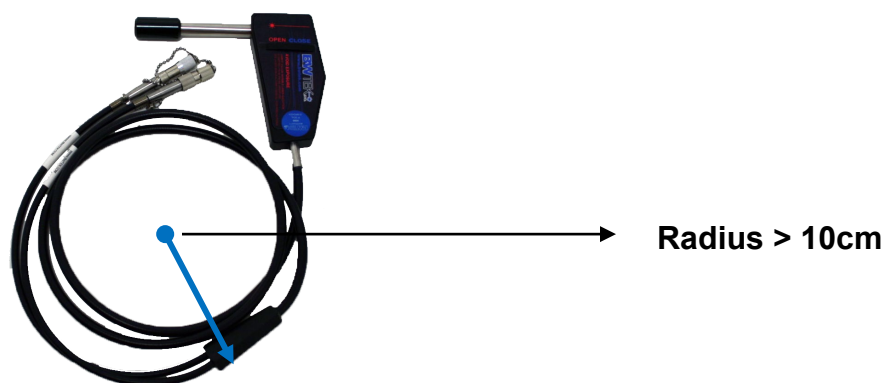
5.1 Exterior

The surfaces of the probe should be kept clean and away from corrosive material.

The housing is not liquid tight. Do not spill liquid over the surface.

Avoid damaging the optic fibers by excessive bending. The recommended minimum bend

radius is 10 cm.



The standard shaft of the probe has a fused silica lens at the tip. The lens surface should be free from any contaminant. It is not an immersion shaft. Do not allow it to touch the sample. In case of contamination, clean with a piece of optic tissue dampened with isopropanol, ethanol, or methanol. The optic surface of other optional shafts can be cleaned in the same way.

5.2 Fiber Optic Tips

The fiber optic terminations of the probe must be protected all the time. Keep the probe connected to the spectrometer during operation and maintenance. The fiber tips of a disconnected probe should be covered by the dust caps, and not be allowed to contact surfaces that could transfer any dirt, dust, or cause scratch to the terminations.

Warning: If there is any reason to suspect a fiber tip is dirty, do not connect it to the spectrometer port until it is properly cleaned. Connecting a contaminated fiber to the port on the instrument can transfer the contaminant to the internal optic surface which can only be cleaned by a service engineer. Furthermore, any foreign materials such as dust particles, debris and tissue materials left on the laser fiber tip can cause the fiber to be burned when the laser is turned on, causing irreversible damage.

Warning: Do not connect or disconnect the excitation fiber to the excitation port while the laser is turned on. Doing so could cause damage to the optic fiber.

When using dust caps to cover up the fiber ports on the spectrometer, or the fiber tips of the probe, make sure the inside surface is clean to avoid transferring contaminants to the ports or fiber tips. Blow the inside surface with compressed air to remove dust.

If a fiber scope is available, it is recommended the fiber tips be inspected before connecting them to the spectrometer ports.

In the event the fiber tips are lightly contaminated, it can be cleaned. There are two methods, a dry cleaning method and a wet cleaning method. If a fiber scope is available, start with a dry cleaning and inspect the result. If this is not effective, then use wet cleaning. If a fiber scope is not available, start with a wet cleaning, followed by a dry cleaning.

Dry Cleaning Method

- Use fiber optic cleaning tool. Expose a fresh surface of the cleaning cloth
- Holding the fiber end perpendicular to the cleaning cloth, twist 90 degrees and then drag down across the exposed cloth applying a small amount of pressure.
- Repeat once. Do not re-use the same section of the cleaning cloth.

Wet Cleaning Method

- Dampen the cleaning cloth with isopropanol or a specially designed cleaning fluid. Do not saturate the cloth.
- Holding the fiber connector perpendicular to the cloth, twist and wipe the end face in the damp area of the cloth several times.
- Repeat once. Do not re-use the same section of the cleaning cloth.

6 Spare Parts

Part Number	Description
840001197	Probe distance regulator